Study Roadmap

Interview Preparation Series • Professional Edition

Master Index

Compiled with LaTeX • June 17, 2026

harsha.somisetty01@gmail.com

In This Edition

Comprehensive, first-principles, interview-focused notes across 11 core SDE/Senior-SDE topics.

Each topic follows the same structure: **Overview → Why it exists → Why FAANG cares → Core concepts → Architecture/diagrams → Real-world examples → Real-life analogy → Memory tricks → Common interview questions → Senior-level discussion → Interview tips → Revision cheat sheet.**

1 Night-Before Cram

`CHEATSHEET.md` — one-page all-topics recall (all 11 topics condensed, ~20 min read). Start here the night before.

2 Magazine Editions (premium PDFs)

Publication-quality, magazine-style PDFs of every topic — custom typography, framed diagrams, infographic callouts and color-coded sections. Each is self-contained (fonts embedded).

Topic	PDF	Topic	PDF
Concurrency	PDF	Distributed Systems	PDF
Operating Systems	PDF	Databases	PDF
Computer Networks	PDF	Performance Eng.	PDF
Cloud & Infra	PDF	Security	PDF
System Design	PDF	Behavioral	PDF
Low-Level Design	PDF		

= flagship hand-crafted edition (custom infographics). Others are generated from the source notes with the same design system. The cheat sheet also has a print PDF. Builders live in `.magazine/`.

Applied-pack magazines:

Pack	PDF	Pack	PDF
DSA · Coding Patterns	PDF	System Design Problems I	PDF
DSA · Data Structures	PDF	System Design Problems II	PDF
DSA · Algorithms	PDF	LLD · Machine Coding	PDF

** AI / ML series** (see `AI-ML/00-INDEX.md`):

Edition	PDF	Edition	PDF
ML Fundamentals	PDF	Transformers & LLMs	PDF
Classic ML Algorithms	PDF	LLM Applications & GenAI	PDF
Deep Learning	PDF	MLOps & ML System Design	PDF

> Preview & edit any edition live: `cd .magazine && python3 serve.py` → http://localhost:8000

3 Topics

#	Topic	File
1	Concurrency & Multithreading	`01-concurrency-multithreading.md`
2	Operating Systems	`02-operating-systems.md`
3	Computer Networks	`03-computer-networks.md`
4	Cloud & Infrastructure	`04-cloud-infrastructure.md`
5	System Design	`05-system-design.md`
6	Low-Level Design (LLD)	`06-low-level-design.md`
6½	Object-Oriented Programming (OOP)	`12-oop-concepts.md`
7	Distributed Systems	`07-distributed-systems.md`
8	Databases	`08-databases.md`
9	Performance Engineering	`09-performance-engineering.md`
10	Security Fundamentals	`10-security-fundamentals.md`
11	Behavioral / Leadership / Googleyness	`11-behavioral-leadership.md`
13	Design Patterns	`13-design-patterns.md`
14	API Design	`14-api-design.md`
15	Caching	`15-caching.md`
16	Message Queues & Streaming	`16-message-queues.md`
17	Testing & Code Quality	`17-testing.md`
18	Observability	`18-observability.md`
19	Git & Dev Workflow	`19-git-version-control.md`
20	Linux & the Command Line	`20-linux-cli.md`
21	Vector Databases & Embeddings	`21-vector-databases.md`

> AI Agents & Tool Use lives in the AI-ML series.

4 Applied Practice Packs

Pack	What's inside
DSA / Coding Round	Pattern recognition + templates, data structures, algorithms & complexity — the #1 interview filter.
System Design Problems	8 fully worked end-to-end designs: Part 1 — TinyURL, Rate Limiter, News Feed, Chat · Part 2 — YouTube, Uber, Distributed Cache, Web Crawler
LLD / Machine-Coding	6 complete runnable Python solutions: LRU Cache, Parking Lot, Rate Limiter, Splitwise, Elevator, Tic-Tac-Toe.

5 How topics connect

```
          OS ◄──────► Concurrency & Multithreading
          │                    │
          ▼                    ▼
     Networks ◄──────► System Design ◄──────► Distributed Systems
          │                    │                      │
          ▼                    ▼                      ▼
    Cloud & Infra ◄──► Performance Eng.         Databases
          │                                           │
          ▼                                           ▼
     Security ◄──────────────────────────────► Low-Level Design

    Behavioral / Leadership wraps ALL of the above (how you communicate them).
```

6 Suggested revision order (last-week-before-interview)

1. System Design (5) + Distributed Systems (7) — highest signal for senior loops
2. Databases (8) + Concurrency (1) — deep technical follow-ups
3. Networks (3) + OS (2) — fundamentals screens
4. LLD (6) + Cloud (4) + Performance (9) + Security (10)
5. Behavioral (11) — review your STAR stories nightly